as being on a B2 or C1 level [6]. Regarding their acquaintance with LLMs, the participants were divided almost equally into regular and irregular LLM users.

Results: 53% of the generated documents received an authenticity rating between five and seven on the Likert scale meaning that they are assessed as rather authentic to very authentic. For comparison, 74% of the real documents got a score in the same range assigned.[7]. There was no noticeable difference in the ratings between regular and irregular users or between the different generation configurations (i.e. between zero- vs. two-shot prompting and greedy vs. more deterministic sampling methods).

In total, participants commented on 451 of 725 document ratings. We have categorized the comments into different positive and negative aspects. A comprehensive table showing these categorizations can be found in the appendix. The most frequent negative points of criticism were the lack of details and too generic contents, unauthentic values for addresses, phone numbers, and other named entities, and repetitiveness regarding words or formulations. Positive comments were predominately about the inclusion of details, a good overall structure, and an authentic writing style with imperfections like spelling or grammar mistakes as well as incomplete sentences or parentheses.

5 Discussion

Overall, the results indicate that the KnoWoGen can generate authentic documents. Still, if the detected issues are resolved, an even higher authenticity is achievable. Moreover, the comments provided insights into which factors humans adduce to identify authentic documents which is valuable to optimize the generation.

Stylistic issues of generated documents, like small formatting issues or repetitive language, have a low impact on the objectives of the KnoWoGen since the produced datasets are intended for evaluating knowledge work support tools and not for Natural Language Processing tasks conducting a deep language analysis. In contrast, content-related issues are more important since content analysis is typical when aiming at supporting knowledge workers. For instance, for information extraction, search or recommendation.

Regarding content-related issues, the lack of details or too generic content were among the most frequently mentioned problems. Documents generated for this experiment were built based on a maximum of one previously generated document. A follow-up experiment

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[6] According to the Common European Framework of Reference for Languages (CEFR) (https://www.europaeischer-referenzrahmen.de/)

[7] Complete score distribution for real documents: (1: 4%, 2: 5%, 3: 6%, 4: 11%, 5: 16%, 6: 28%, 7: 31%) and for generated ones: (1: 8%, 2: 14%, 3: 16%, 4: 9%, 5: 15%, 6: 17%, 7: 20%). The KL divergence of the two distributions is 15.91%. A visualization can be found at the KnoWoGen website: https://purl.archive.org/knowogen/document_authenticity_experiment

could examine whether the situation improves when using documents generated based on more previously generated documents. Besides, the prompts can be enhanced to encourage a more specific focus and more details which can likewise be achieved by a preceding step in which the focus aspects are generated. In some cases, aspects to focus on can also be predetermined. However, this would decrease the documents' diversity.

Another typical problem was the generation of unauthentic names, addresses, or phone numbers that were filled in since this information was not part of the prompt. This could be solved by extracting such kinds of information for different document types from several generated documents and providing more authentic values explicitly in future prompts.

Furthermore, the documents showed that the role of the author was not clear during the generation. In a few documents, the intended author given in the prompt did refer to themselves in the third person. Thus, in the future, the authorship must be made clearer including instructions stating that the author should refer to themselves in the first person.

Multiple other issues can be potentially solved by including more general instructions in the prompt, e.g., how to refer to colleagues or externals, keeping the document concise, or adding variations. Additionally, instructions could encourage characteristics that the experiment revealed as being particularly authentic, e.g., a few spelling or grammatical errors could make the resulting documents more realistic.

In summary, the experiment showed that the KnoWoGen can generate documents perceived as authentic. Moreover, the results indicate which aspects were most influential for the participants' judgments. This information can be very beneficial for optimizing the authenticity of documents even more. For future experiments, examining other document types, multiple related documents, and inter-document diversity would be interesting. In contrast, examining the knowledge graph or the simulation setting is not as meaningful as analyzing the generated documents as they reflect, by design, the customizable configuration.

6 Conclusion

In this paper, we proposed KnoWoGen, a knowledge work dataset generator that can be tailored to different evaluation needs. It targets common problems of existing data collections, like incompleteness, lack of background information, and limited applicability. The generator simulates multiple knowledge workers collaborating and completing tasks. As an outcome of most actions, documents are generated by prompting an LLM with the necessary contextual information. Our experiment showed that this approach is promising and indicated some optimization potential. In future works, document and task interdependencies could be increased, diversity in larger document collections could be analyzed and optimized, and synthesized documents could be used to influence the simulation more, e.g., by creating additional tasks derived from previously generated documents.

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